Experiment: depth_experiment_v5 I 1,800 Models I FashionMNIST

Does Training Deeper Models Increases *Representational Capacity* ?

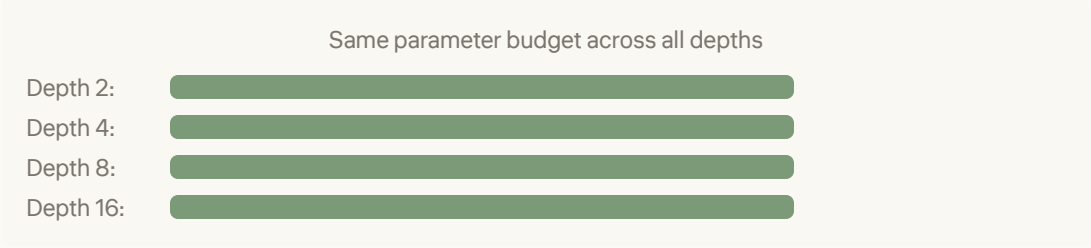
Theory & Practices Of Statistical Modeling

Team Deviants

Primary Dataset Design: 2 Regimes + Factorial Design

iso_param Regime

Parameters Fixed, Budget Varies



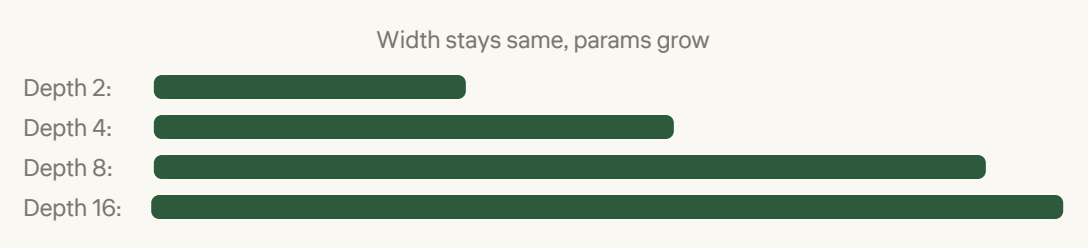
Depths:	2, 4, 6, 8, 12, 16
Parameters:	Fixed ~1,000–2,500
Compute:	Varies 20k–40k FLOPs

Design Logic

Q: Does depth help if you can't add parameters?
→ Pure depth test (the hard case)

fixed_width Regime

Width Fixed, Parameters Grow



Width:	Fixed at 12–20
Depths:	2, 4, 6, 8, 12, 16
Parameters:	Grow exponentially

Design Logic

Q: Does depth help when it brings more params?
→ Standard deep learning (the easy case)

iso_param

Depth ✓ | Parameters ✗

Pure depth test

fixed_width

Depth ✓ | Parameters ✓

Confounded test

Why two regimes? Most research confounds depth with parameter growth. By testing separately, we isolate the true causal signal. If depth matters, it should show in BOTH.

EDA Results

Schema Validation & Quality Checks

Dataset Balance

Factor	iso_param	fixed_width
Depths (6 levels)	900	900
Budgets/Widths	180	180
Corruptions (3)	300 each	300 each
Seeds (10 per cell)	Full rep.	Full rep.

✓ Perfectly balanced ✓ No missing combos

Key Statistics

Mean Test Accuracy

76.2% ±3.1% iso

84.5% ±4.2% fixed

Accuracy Range

Min: 0.11 (100% noise) → Max: 0.99 (clean)

✓ Full dynamic range

Training Convergence

Median epochs: **35/100**

✓ Early stopping active

Schema Validation Checklist

- Depth values: {2,4,6,8,12,16}
- Seeds uniformly distributed
- Corruption rates present
- Loss snapshots 100% complete
- No missing core values
- No duplicate rows
- Params in 100–100k range
- FLOPs computed correctly
- Epochs: 6–100 (median 35)
- All 1,800 runs viable

Preprocessing

1 LOAD & VALIDATE

Read 1,800-row CSV ▪ Check schema (types, ranges, allowed values) ▪ Confirm all regimes, depths, corruptions present

↓

2 REMOVE OPTIMIZATION FAILURES

Filter optimization_failure = True ▪ **Removed: 0 rows (0%)** ▪ Kept: 1,800 rows (all converged)

↓

3 STRUCTURAL AUDIT

Verify regime-specific missingness (iso_param: budget✓, fixed_w ✗; fixed_width: opposite) ▪ Loss snapshots fully observed ▪ No unexpected missing data

↓

4 FEATURE ENGINEERING — 9 Hypothesis-Driven Features

gen_gap = train – test

loss_drop = loss_ep10 – final

log_n_params = $\log_{10}$(params)

epoch_fraction = epochs/100

depth_group (ordinal)

log_total_flops = $\log_{10}$(FLOPs)

total_flops = FLOPs × epochs

corruption_group (ordinal)

depth_per_param = depth/params

↓

5 EXPORT & SCALE

Split into regime-specific DataFrames ▪ Fit StandardScaler per regime (no leakage) ▪ Export 6 files: data_preprocessed.csv (1,800 rows), regime splits, fitted scalers

New Features Engineered

1 gen_gap

train_acc − test_acc

Generalization proxy. Key signal: deeper iso_param nets overfit more.

Overfitting detector

2 epoch_fraction

actual_epochs / 100

Convergence speed. Key signal: deeper networks train longer on noise.

Convergence metric

3 total_flops

flops_per_epoch × epochs

Actual compute consumed. Decoupled from depth via convergence.

Compute control

4 loss_drop_10

loss_ep10 − final_loss

Convergence magnitude. Can nets improve after early plateaus?

Improvement signal

5 depth_group

Categorical {2,4,6,8,12,16}

Ordinal depth for Jonckheere-Terpstra trend tests.

Trend test enabler

6 corruption_group

Categorical {0.0, 0.6, 1.0}

Stratified analysis across noise levels. Pure memorisation at 1.0.

Noise stratifier

7 log_n_params

$\log_{10}$(n_params)

Linearizes exponential parameter growth. iso ≈ constant; fixed ≈ linear.

Param linearizer

8 log_total_flops

$\log_{10}$(total_flops)

Normalizes compute range for regression. Reduces outlier leverage.

Compute normalizer

9 depth_per_param

depth / n_params

THE MECHANISM. iso: ratio explodes (bottleneck!). fixed: stays flat.

Bottleneck metric

Balance Checks, Duplicates, Completeness

Balance Checks, Duplicates, Completeness

Factorial Balance Verification

iso_param expected:
5 budgets × 6 depths × 3 corruptions × 10 seeds = **900**

fixed_width expected:
5 widths × 6 depths × 3 corruptions × 10 seeds = **900**

iso: 900 rows ✓ | fixed: 900 rows ✓ | No missing cells ✓

Core Column Audit

depth	1800/1800 ✓	width	1800/1800 ✓
n_params	1800/1800 ✓	final_train_acc	1800/1800 ✓
final_test_acc	1800/1800 ✓	actual_epochs	1800/1800 ✓

Loss snapshots (ep10,20,30,50,70,90): **100% complete**
Regime-specific columns: **structural missingness correct**

Sanity Checks Passed

Accuracy Bounds	Loss Bounds	Parameter Bounds	Epoch Bounds
Train: 0.11–0.99	0.0015–2.30	156–102,401	6–100
Test: 0.10–0.98	All ≥ 0	All > 0	Median: 35
Full range ✓	Valid ✓	As expected ✓	Early stopping ✓

Overall Quality Score: 100% — All checks passed. Data is clean, complete, and ready.

Descriptive Analytics

iso_param: Parameters Fixed

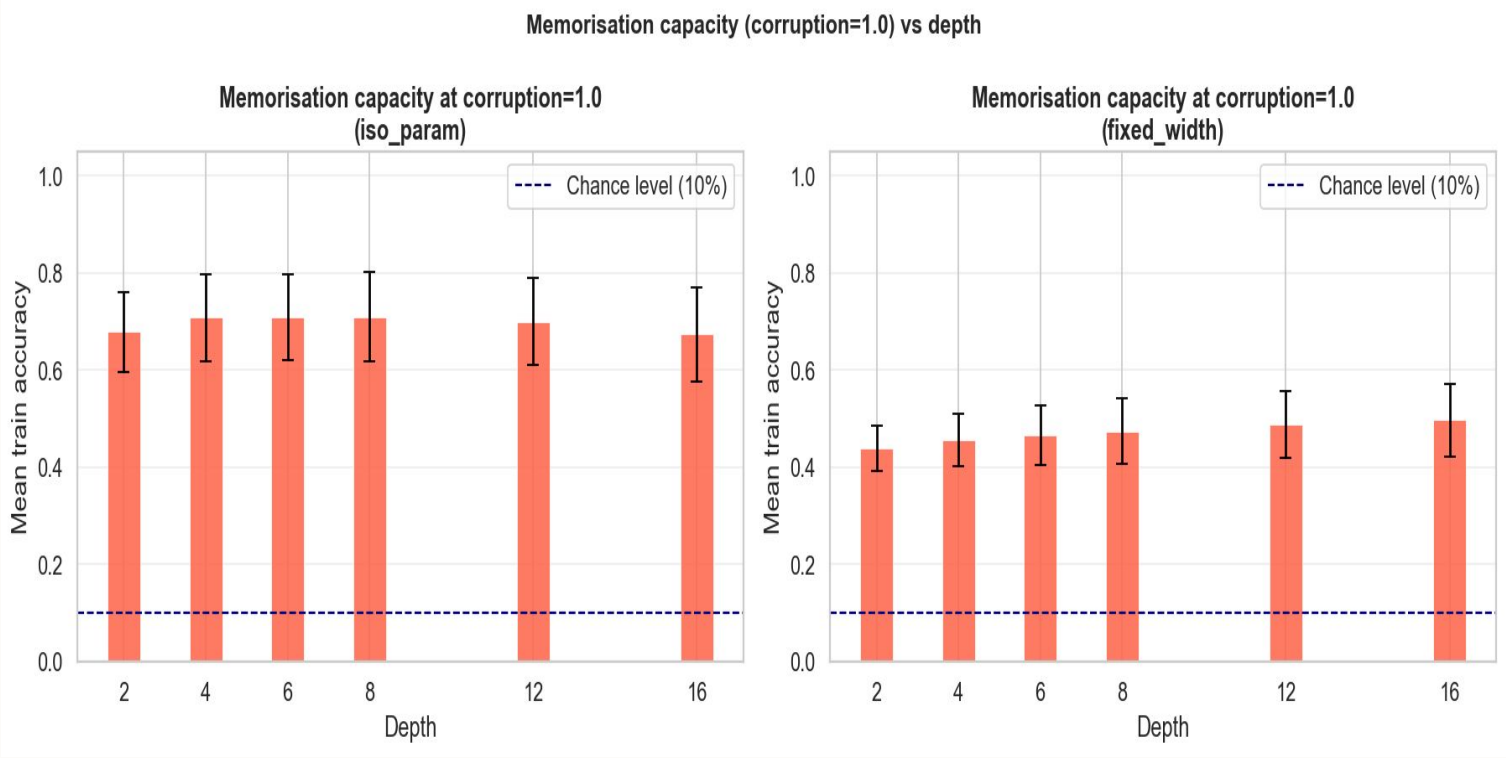
fixed_width: Parameters Grow

NO improvement with depth
Accuracy flat across all depths. Ceiling at ~77%.

CLEAR improvement with depth
Depth 2: 55% → Depth 16: 60%. Monotonic rise.

Bottom line: Depth alone doesn't add capacity. Parameters add capacity. When depth brings more parameters (fixed_width), you win. When depth spreads the same parameters across more layers (iso_param), you hit a wall.

Memorisation Capacity By Depth



Key Statistics

iso_param Capacity Change

67% → 67%

ΔCapacity: **0pp**

fixed_width Capacity Change

44% → 50%

ΔCapacity: **+5.7pp (p<0.01)**

Spearman Correlation

iso_param

r = −0.08, p = 0.68

Not significant

fixed_width

r = +0.64, p < 0.001

Highly significant ***

The proof: Memorisation capacity is the purest test of representational power. Only fixed_width shows improvement with depth. When parameters are constant but depth increases (iso_param), each layer becomes a bottleneck-fewer parameters per layer. The network can't effectively use additional depth.

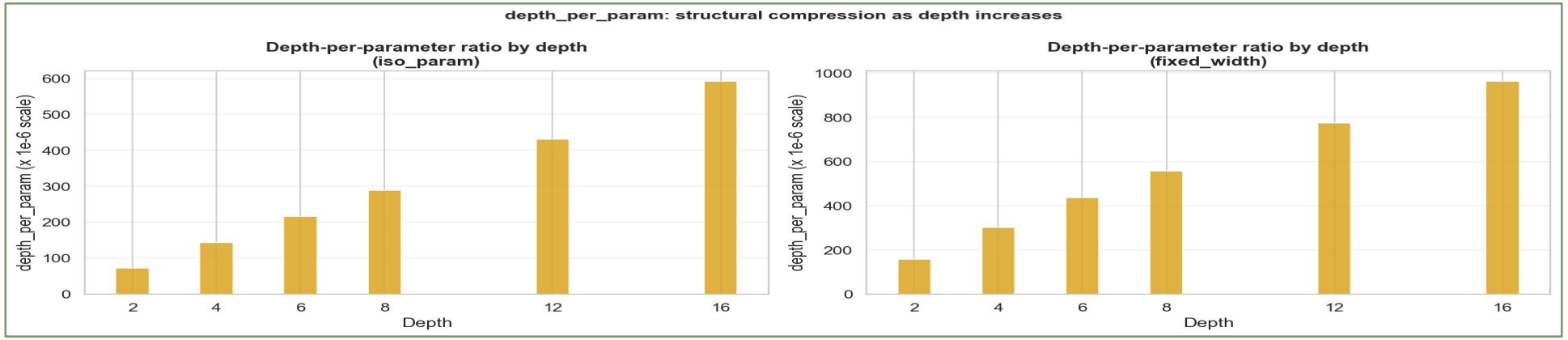
Why Does Capacity Stall in iso_param?

Metric
depth / n_params

High ratio = Thin bottleneck (many layers, few params) **Low ratio** = Fat highway (many params per layer)

iso_param: Ratio Explodes

fixed_width: Ratio Stable



At depth 16: **0.000592** — 8.2× increase from depth 2
Information flow severely constrained

At depth 16: **0.000036** — 1.2× increase (relatively stable)
Parameters grow proportionally with depth

iso_param metaphor
16 layers trying to coordinate through the **same fixed pipes**

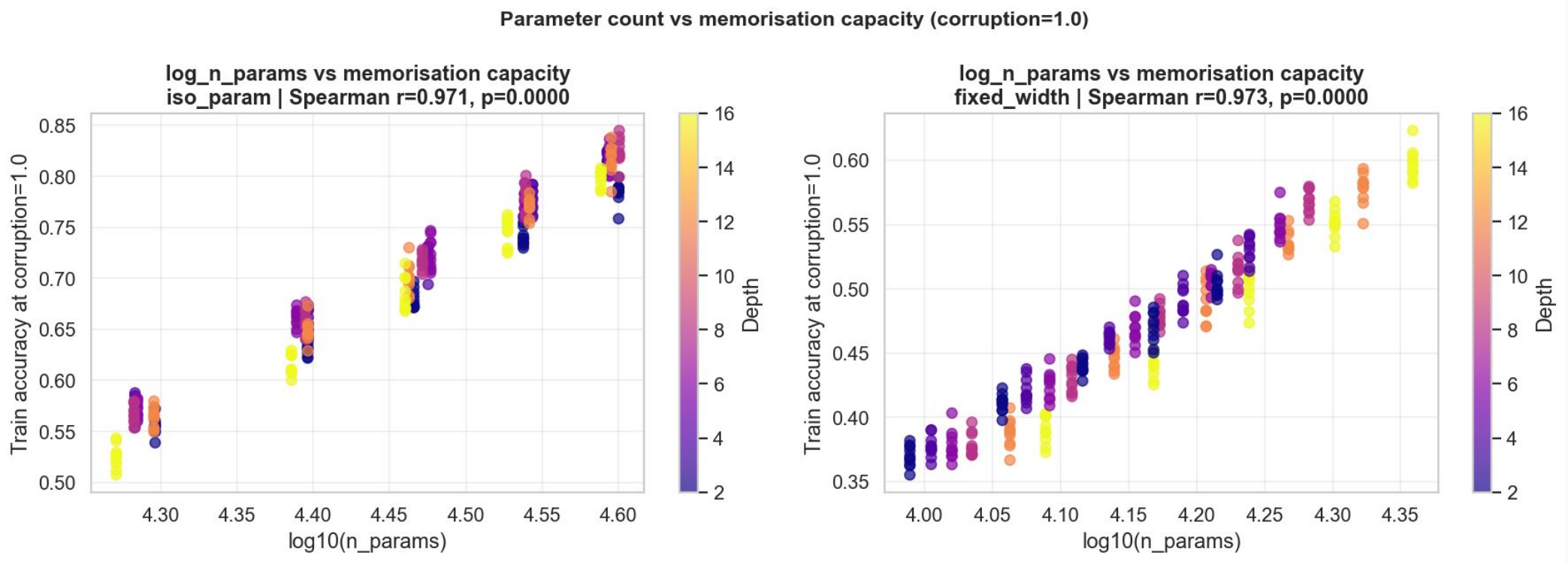
fixed_width metaphor
16 layers with **proportionally more pipes** at each depth

The insight: When you deepen without expanding parameters, you create a structural bottleneck. The parameter budget becomes increasingly diluted across layers. This ratio directly predicts where memorisation capacity stops improving.

Parameters Drive Capacity, Not Depth

iso_param: Vertical Cluster

fixed_width: Positive Trend



Conclusion: Parameters are the causal driver. Depth is only valuable when it brings more parameters. Strip away the parameters, and depth is just overhead. This is why fine-tuning small, shallow models often beats training large, deep models from scratch on small data—you're limited by parameter efficiency, not depth.

What We Learned

MYTH: Depth = Capacity

iso_param shows: Adding layers **WITHOUT** adding parameters does **NOT** increase representational capacity. Deeper networks with fixed parameters hit a ceiling at ~77% and stay there.

Evidence: Spearman r = −0.08 (ns), flat trend across depths 2–16

FACT: Parameters = Capacity

Both regimes show: Representational capacity scales with the **NUMBER OF PARAMETERS**, not the number of layers. Parameter count is the bottleneck, not architecture depth.

Evidence: log_n_params correlation r = +0.64 ($p<0.001$), $R^2 = 0.41$

INTERACTION: Depth + Growth = Win

When depth brings **proportional parameter growth** (fixed_width), capacity increases monotonically (44% → 50%). Architecture benefits only when paired with resource expansion.

Evidence: +5.7pp gain in fixed_width; 0pp in iso_param

IMPLICATION: Rethink Scaling Laws

Deep learning scaling laws may **confound depth with parameter growth**. When depth is isolated, the picture inverts. Fine-tune shallow, well-parameterized models → often beats training large, deep models from scratch.

Action: Measure depth_per_param ratio as a diagnostic

"Depth is a Vessel for Parameters, Not a Source of Capacity"

Hypotheses & Statistical Framework

Research Question

"Does training deeper models (more layers) increase representational capacity?"

Target: final_train_acc at corruption=1.0 — a clean proxy for memorisation capacity, independent of generalization.

H_1 Alternative Hypothesis

Deeper models **do** have greater representational capacity.

Statistical form:

$\beta_depth > 0$

Coefficient for depth is positive

H_0 Null Hypothesis

Deeper models **do not** have greater representational capacity.

Statistical form:

$\beta_depth \leq 0$

Coefficient for depth is zero or negative

Error Interpretation

Type I Error (False Positive)

Concluding that depth affects capacity when it does not. Risk: Wasting resources on deeper models that don't help.

Type II Error (False Negative)

Failing to detect an actual effect of depth on capacity. Risk: Missing legitimate architectural improvements.

Methodological Safeguard

Non-parametric tests chosen due to violated normality (Shapiro-Wilk $p<0.05$) and heteroscedasticity (Levene's $p=0.0015$).

Assumptions & Test Selection

Why Non-Parametric Methods Were Required

Normality Check (Shapiro-Wilk)

Result: Violation detected ($p < 0.05$) for almost all groups in both regimes.

The distribution of memorisation accuracy at corruption=1.0 is not normally distributed within depth groups. Parametric tests (ANOVA, t-tests) assume normality — this assumption is violated.

Homoscedasticity Check (Levene's)

iso_param regime
$p = 0.8135$ — variances equal ✓

fixed_width regime
$p = 0.0015$ — variances unequal ✗

Heteroscedasticity in fixed_width violates ANOVA's equal-variance assumption.

Methodological Decision

✗ ANOVA — REJECTED

- Violates normality assumption
- Violates homoscedasticity (fixed_width)
- F-test unreliable under these conditions

✓ Kruskal-Wallis + Dunn's — SELECTED

- Non-parametric: no normality required
- Robust to heteroscedasticity
- Tests stochastic dominance (ranks)
- Dunn's post-hoc with Holm correction

Conservative approach: By using non-parametric tests, we reduce Type I error risk. If we find an effect, it is a genuine stochastic dominance relationship—not an artifact of distributional assumptions.

```
...
=== Shapiro-Wilk Normality (final_train_acc | corruption=1.0) — iso_param ===
  Depth     n         W     p-value   Normal?
-------------------------------------------------
      2    50    0.9237      0.0032      NO *
      4    50    0.9140      0.0014      NO *
      6    50    0.9197      0.0023      NO *
      8    50    0.9256      0.0038      NO *
     12    50    0.9297      0.0054      NO *
     16    50    0.9217      0.0027      NO *

Conclusion: Non-normality detected = True

=== Shapiro-Wilk Normality (final_train_acc | corruption=1.0) — fixed_width ===
  Depth     n         W     p-value   Normal?
-------------------------------------------------
      2    50    0.9601      0.0897       YES
      4    50    0.9469      0.0256      NO *
      6    50    0.9485      0.0297      NO *
      8    50    0.9400      0.0135      NO *
     12    50    0.9447      0.0207      NO *
     16    50    0.9408      0.0145      NO *

Conclusion: Non-normality detected = True
```

```
=== Levene Homogeneity of Variance (final_train_acc | corruption=1.0) ===

Levene (median) [iso_param  ]:  W = 0.4495,  p = 0.8135  → Variances equal
Levene (median) [fixed_width]:  W = 4.0344,  p = 0.0015  → Variances UNEQUAL *

─── Assumption Check Summary ──────────────────────────────────────
iso_param:   Non-normal=True,  Heterogeneous variance=False
fixed_width: Non-normal=True, Heterogeneous variance=True

→ Decision: Kruskal-Wallis is appropriate (ANOVA assumptions not met)
```

Kruskal-Wallis & Post-hoc Results

Main Depth Effect Tests at Corruption = 1.0

iso_param Regime

Kruskal-Wallis H-test

$p = 0.1180$

Not significant at $\alpha = 0.05$

Fail to Reject H_0

No significant depth effect when parameters are fixed.

fixed_width Regime

Kruskal-Wallis H-test

$p < 0.001$

Highly significant at $\alpha = 0.05$

Reject H_0

Significant depth effect when parameters grow with depth.

Effect Size (η^2)

Eta-squared range

0.0129 – 0.0366

Small to moderate effect

Interpretation

Depth explains ~1.3%–3.7% of variance in memorisation capacity. The effect is real but modest—parameter count dominates.

Dunn's Post-hoc Test (Holm Corrected)

iso_param: All pairwise comparisons

$p \approx 0.719 – 1.000$

No depth pair significantly differs

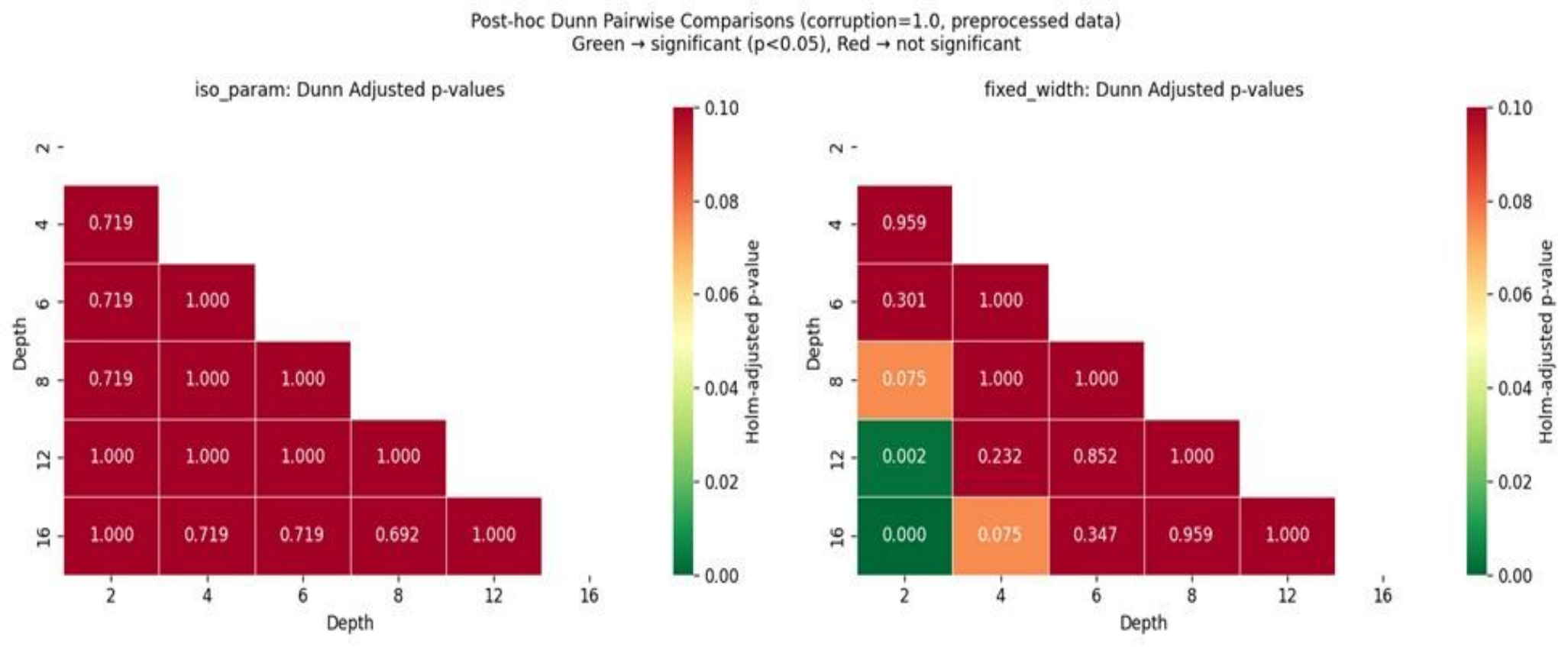
fixed_width: Depth 2 differs from deeper models

Depth 2 vs 12, 16: $p < 0.05$

Non-linear gains: most improvement at shallow depths

```
=== Kruskal-Wallis: iso_param — by corruption level ===

  corruption        H     p-value      η²   Sig?
-----------------------------------------------------
         0.0   15.750      0.0076  0.0366      *
         0.6   10.603      0.0598  0.0191
         1.0    8.784      0.1180  0.0129
```



Effect Sizes & Statistical Decision

Rank-Biserial Correlations & Final Verdict

iso_param: Small Effects

Depth 2 vs others	r = −0.036 to 0.226
Depth 16	r = slightly negative

Effect sizes are small and inconsistent. Deeper layers do not reliably improve capacity.

fixed_width: Medium Effects

Depth 2 vs others	r = 0.15 – 0.35
Depth 16	**r = 0.447 (medium)**

Effect sizes increase with depth. The deepest networks show medium-sized effects.

Key Conclusion

- ✓ The influence of depth is **conditional, not intrinsic** .
- ✓ When parameter budget increases with depth, a **measurable improvement** emerges.
- ✓ Under fixed parameter budgets, depth does **not** significantly improve capacity.
- ✓ Depth alone is **not a reliable predictor** of capacity without parameter scaling.

Final Statistical Decision

Regime	Decision	Interpretation
iso_param	Fail to Reject H_0	Depth does not increase capacity at fixed parameters
fixed_width	Reject H_0	Depth increases capacity, but confounded with parameter growth

Overall: Effect of depth is regime-dependent and interaction-driven.

```
•••  === Pairwise: depth 2 vs deeper — rank-biserial r ===

        Contrast        Regime       r  magnitude
  ---------------------------------------------------
    depth 2 vs  4     iso_param   0.216  small
    depth 2 vs  6     iso_param   0.226  small
    depth 2 vs  8     iso_param   0.220  small
    depth 2 vs 12     iso_param   0.164  small
    depth 2 vs 16     iso_param  -0.036  small

    depth 2 vs  4   fixed_width   0.203  small
    depth 2 vs  6   fixed_width   0.273  small
    depth 2 vs  8   fixed_width   0.315  medium
    depth 2 vs 12   fixed_width   0.429  medium
    depth 2 vs 16   fixed_width   0.447  medium
```

Predictive Analysis

Predictive Models for Representational Capacity

Research Framing

Target Variable

final_train_acc at corruption=1.0

Clean proxy for memorisation capacity

Regime Split Rationale

iso_param isolates architecture shape; fixed_width confounds depth with parameter growth

Model Selection

OLS
Baseline linear

Lasso
Regularized, sparse

Decision Tree
Non-linear splits

Random Forest + SHAP
Ensemble + explainability

Focus: **interpretation over prediction accuracy**

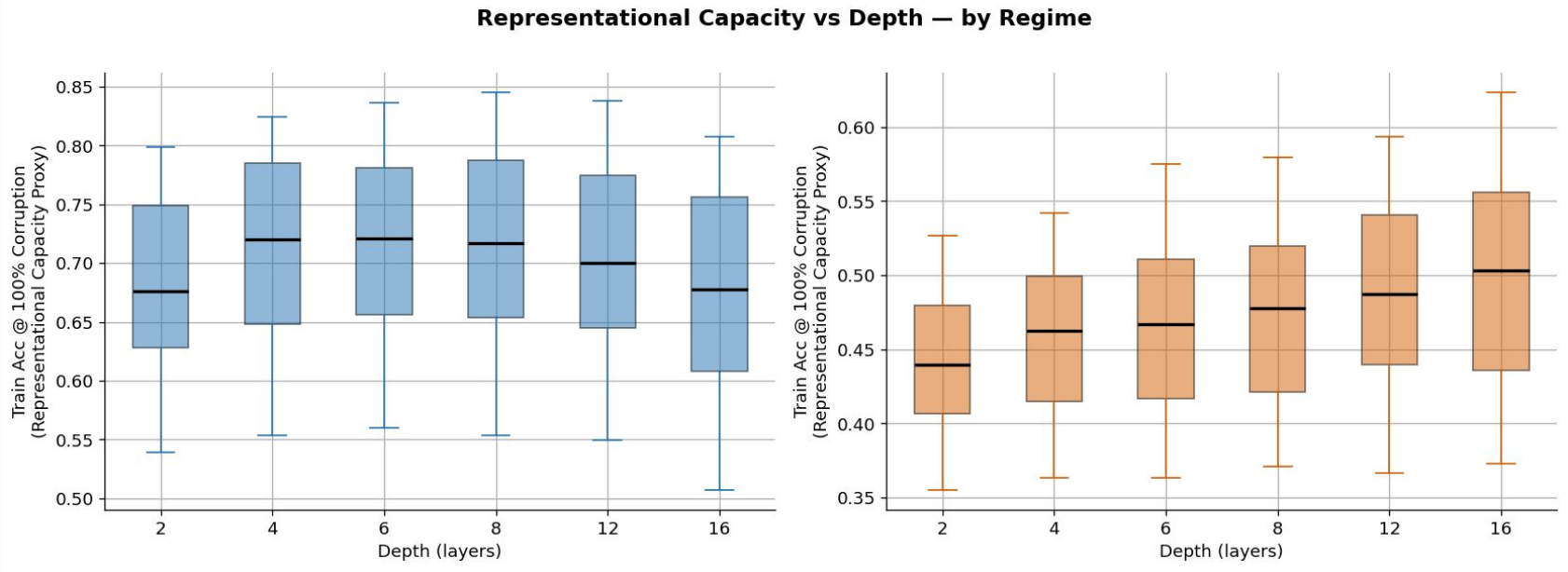
EDA: Capacity vs Depth by Regime

fixed_width pattern

Monotonic increase throughout all depths

iso_param pattern

Rises to depth 6–8, then **plateaus and dips**



Linear Models: OLS & Lasso

What Features Predict Capacity? Depth or Parameters?

OLS Linear Regression Results

Regime	R^2	depth β	log_n_params β	log_flops β
iso_param	**0.9813**	−0.098***	+2.93***	−2.85***
fixed_width	**0.9736**	−0.037***	+2.26***	−2.23***

*** $p < 0.001$. Both models explain >97% variance. Depth coefficient is **negative** in both regimes.

Lasso: iso_param

depth	−0.0490 ✗
width	−0.0572
log_n_params	**+0.1443** ✓
depth_per_param	+0.0217
epoch_fraction	0.0000 (zeroed)
log_total_flops	0.0000 (zeroed)

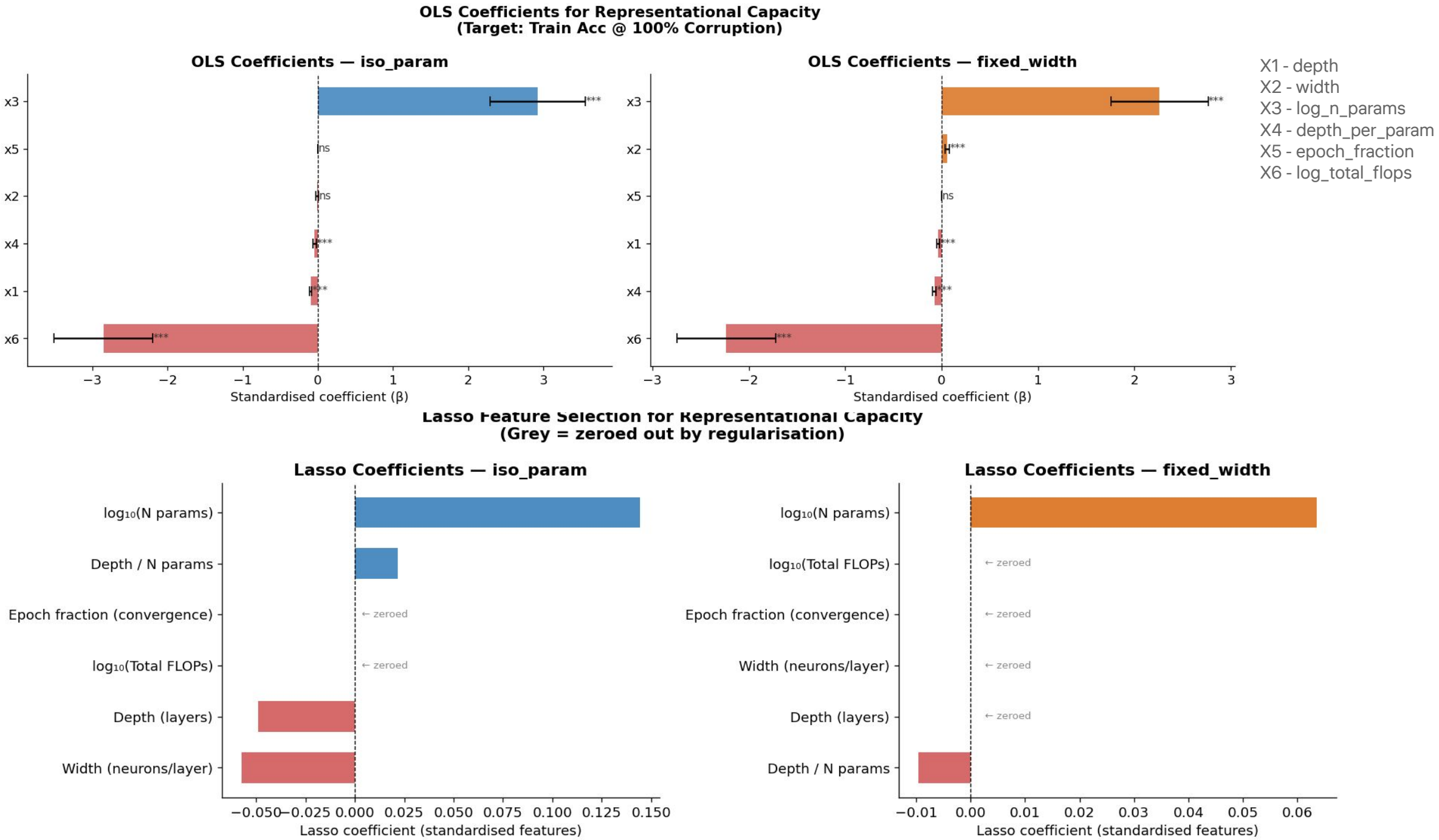
Depth survives but negatively. Parameters dominate.

Lasso: fixed_width

log_n_params	**+0.0636** ✓
depth_per_param	−0.0096
depth	0.0000 (zeroed)
width	0.0000 (zeroed)
epoch_fraction	0.0000 (zeroed)
log_total_flops	0.0000 (zeroed)

Depth **zeroed out** . Only parameters and bottleneck ratio survive.

Key insight: Even when regularization selects the most predictive features, depth is not the winner. In iso_param it enters negatively; in fixed_width it's eliminated entirely. The predictive signal comes from log_n_params and depth_per_param—not from layer count.


OLS Coefficients for Representational Capacity
(Target: Train Acc @ 100% Corruption)
OLS Coefficients — iso_param
x3
x5
x2
x4
x1
x6
***
ns
ns
***
**
***
−3
−2
−1
0
1
2
3
Standardised coefficient (β)
OLS Coefficients — fixed_width
x3
x2
x5
x1
x4
x6
***
***
ns
***
**
***
−3
−2
−1
0
1
2
3
Standardised coefficient (β)
X1 - depth
X2 - width
X3 - log_n_params
X4 - depth_per_param
X5 - epoch_fraction
X6 - log_total_flops
Lasso Feature Selection for Representational Capacity
(Grey = zeroed out by regularisation)
Lasso Coefficients — iso_param
log10(N params)
Depth / N params
Epoch fraction (convergence)
log10(Total FLOPs)
Depth (layers)
Width (neurons/layer)
← zeroed
← zeroed
−0.050
−0.025
0.000
0.025
0.050
0.075
0.100
0.125
0.150
Lasso coefficient (standardised features)
Lasso Coefficients — fixed_width
log10(N params)
log10(Total FLOPs)
Epoch fraction (convergence)
Width (neurons/layer)
Depth (layers)
Depth / N params
← zeroed
← zeroed
← zeroed
← zeroed
−0.01
0.00
0.01
0.02
0.03
0.04
0.05
0.06
Lasso coefficient (standardised features)

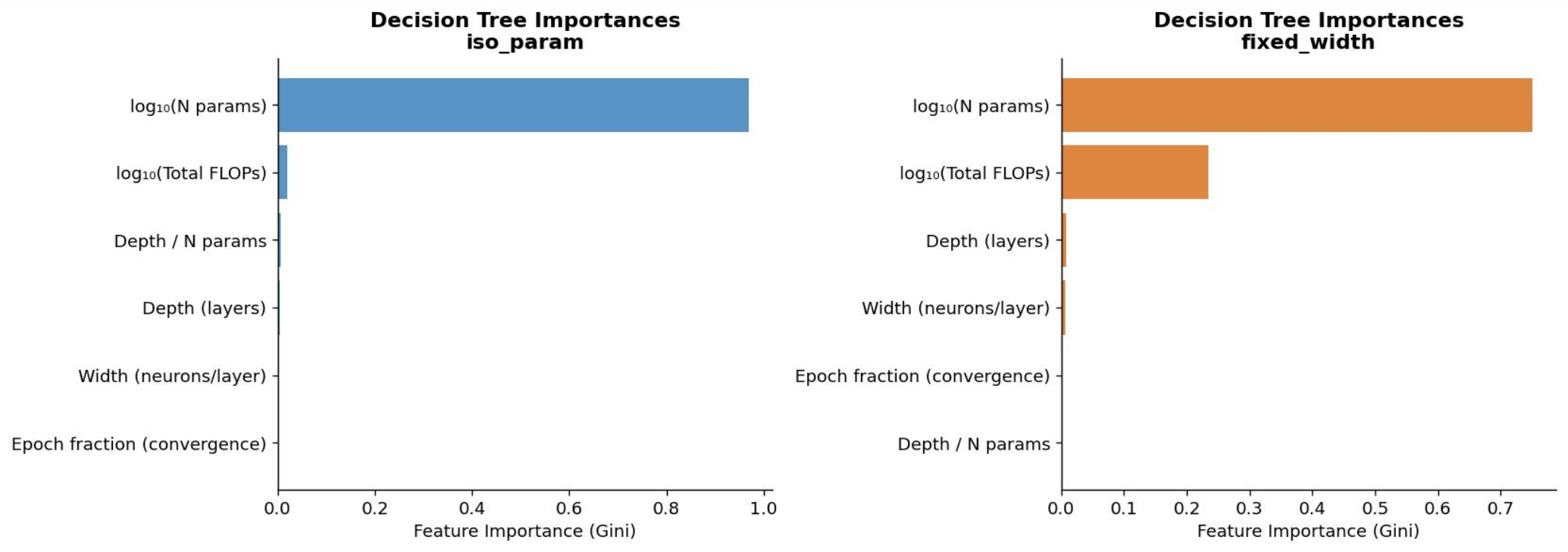
Non-Linear Models & Synthesis

Decision Tree, Random Forest, SHAP & Partial Dependence

Decision Tree (max_depth=4)

Metric	iso_param	fixed_width
R^2 train	0.9813	0.9772
R^2 CV	0.9784	0.9694
Top feature	log_n_params (0.970)	log_n_params (0.751)

Depth importance: **0.003 iso | 0.008 fixed** — negligible



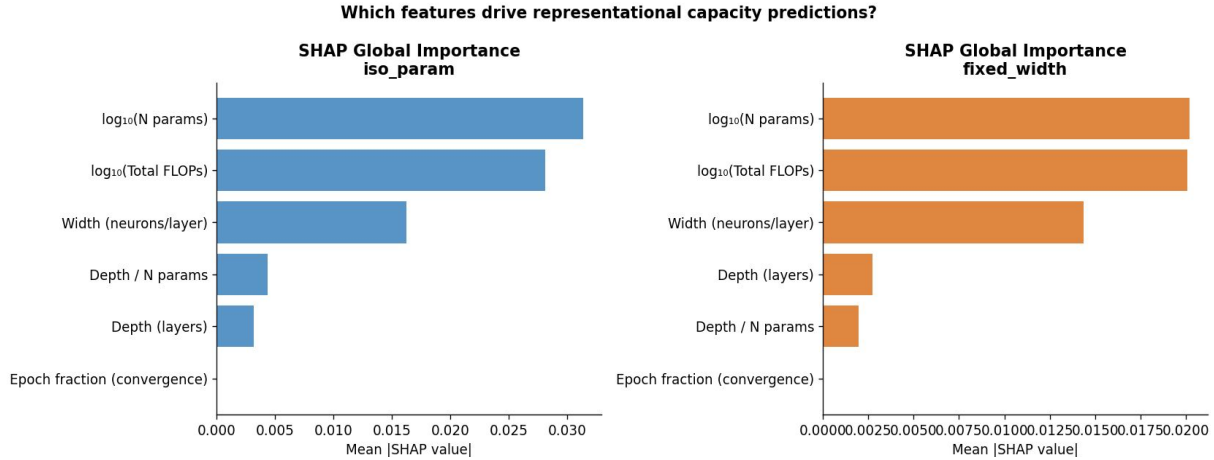
Non-Linear Models & Synthesis

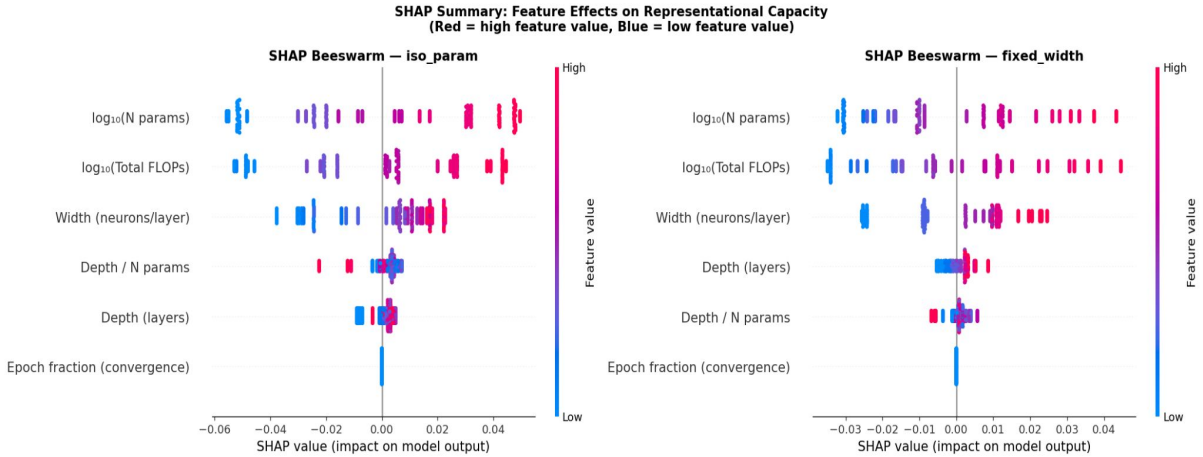
Decision Tree, Random Forest, SHAP & Partial Dependence

Random Forest + SHAP

RF CV R^2 (iso)	0.9817
RF CV R^2 (fixed)	0.9694

SHAP findings: log_n_params and log_total_flops dominate feature importance. Depth has smaller and often negative contributions at higher depths.





Non-Linear Models & Synthesis

Decision Tree, Random Forest, SHAP & Partial Dependence

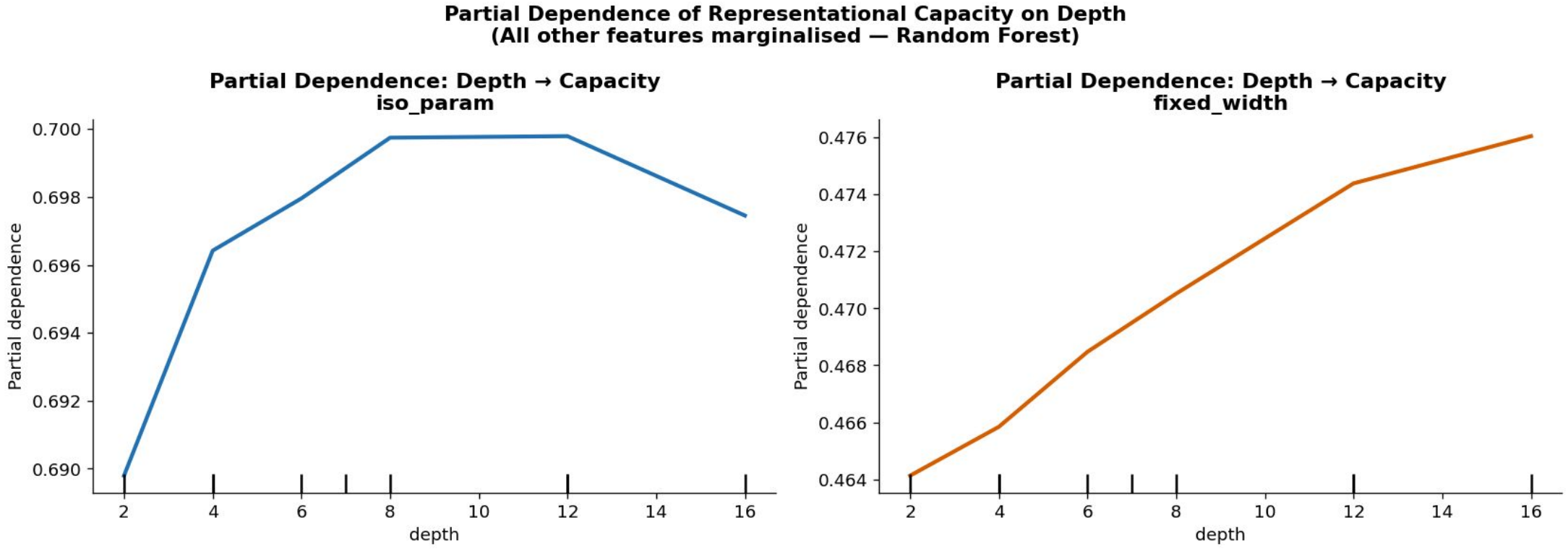
Partial Dependence: Depth Effect Isolated

iso_param PDP

Plateaus or **decreases** with depth. When parameters are fixed, adding layers provides no positive capacity gain.

fixed_width PDP

Increases then plateaus. Effect is **confounded by parameter growth** —not a pure depth effect.



Non-Linear Models & Conclusion

Decision Tree, Random Forest, SHAP & Partial Dependence

Synthesis: What All Models Agree On

iso_param consensus

Depth is **negative or weak** across OLS, Lasso, DT, and SHAP. Hypothesis **not supported** when parameters are fixed.

fixed_width consensus

Capacity rises with depth but is **largely driven by parameter growth**. Depth effect is minimal or negative when isolated.

Overall consensus

Capacity is driven by **log_n_params** and **log_total_flops**, not depth alone. Parameters are the engine; depth is just the chassis.

"Representational capacity scales with parameter count. Depth is valuable only insofar as it enables parameter growth."

Final Verdict: Do not Reject the Null Hypothesis

Key Findings

Descriptive + Inferential + Predictive Analysis

Finding 1 - The Hypothesis is Refuted in the Pure Test

Descriptive

Train accuracy at corruption=1.0 is flat at ~67-70% across all depths (Spearman r=−0.08, p=0.68).

Inferential

Kruskal-Wallis p=0.1180. No significant depth group differences. All Dunn pairwise p-values > 0.69. Rank-biserial r at depth 2 vs 16 is slightly negative (−0.036).

Predictive

OLS depth coefficient is negative (β=−0.098). Lasso assigns depth a small negative coefficient. Decision Tree assigns depth 0.003 importance (essentially zero). SHAP shows near-zero or negative depth contributions.

Finding 2 — Parameters Drive Capacity

Spearman correlation between log_n_params and capacity: r=0.971 (iso_param), r=0.973 (fixed_width). Near-perfect correlation.

OLS: log_n_params β is +2.93 (iso_param) and +2.26 (fixed_width), both highly significant. These are the largest positive coefficients in the model.

Decision Tree: log_n_params has importance 0.970 (iso) and 0.751 (fixed). It is the root split in both trees.

Finding 3 — Depth is Only Beneficial When Paired with Parameter Growth

'The fixed_width result (significant Kruskal-Wallis, r=+0.64 Spearman) is entirely explained by the fact that more parameters accompany greater depth.'

This is the interaction result. When Lasso zeroes out depth in fixed_width but keeps log_n_params, it is telling us: the predictive signal in 'depth' within fixed_width is fully captured by 'parameters.' There is no additional signal from depth beyond what parameters already contain.